

HW4

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In the following, let $\mathbb{F}$ be a field, which satisfies $\text{char}\mathbb{F} = 0$ and $\overline{\mathbb{F}} = \mathbb{F}$ in Exercises 3 and 4.

Exercise 4.1

Let L be a nilpotent Lie algebra over $\mathbb{F}$. Prove that the Killing form of L is identically zero.

Proof. From Engel's Theorem, there is a basis of L , say $\{e_1, \dots, e_n\}$, in which every element of $\mathfrak{gl}(L)$ is represented strictly upper triangular. $\square$

Proof. $\text{ad}(L)$ is a Lie subalgebra of $\mathfrak{gl}(L)$ in which every element is nilpotent. It follows from Engel's Theorem that there is a basis $\{e_1, \dots, e_n\}$ of L in which every element of $\text{ad}(L)$ is represented by a strictly upper triangular matrix. Take any $x, y \in L$. Suppose $\text{ad}(x), \text{ad}(y)$ are represented by strictly upper triangular matrix $A = (a_{ij}), B = (b_{ij})$ respectively. Then for each $i \geq k$, we have $a_{ik} = 0$, and for each $i \leq k$, $b_{ki} = 0$, thus

$$\kappa(x, y) = \text{tr}(AB) = \sum_{i=1}^n \sum_{k=1}^n a_{ik} b_{ki} = 0$$

The Killing form of L is identically zero since x, y are arbitrary. $\square$

Exercise 4.2

Classify the 2-dimensional Lie algebras over $\mathbb{F}$, and compute their Killing form.

Proof. • The first kind of 2-dimensional Lie algebra over $\mathbb{F}$ consists of abelian Lie algebras, which are unique in the meaning of Lie isomorphism. Since they are specifically nilpotent, it follows from Exercise 4.1 that their Killing form are identically zero.

- Now suppose L is a non-abelian Lie algebra of dimension 2 over the field $\mathbb{F}$. The derived algebra of L cannot be more than 1-dimension since $\{x, y\}$ is a basis of L , and L' is spanned by $[x, y]$. The other hand, L' is nonzero otherwise L is abelian. Therefore L' must be 1-dimensional. Take a nonzero element $x \in L'$ and extend it to a basis $\{x, \tilde{y}\}$ of L . Then $[x, \tilde{y}] \in L'$ and is nonzero. There is a nonzero scalar $\alpha \in F$ such that $[x, \tilde{y}] = \alpha x$. By replacing $\tilde{y}$ by $y := \alpha^{-1}\tilde{y}$, we get

$$[x, y] = x$$

The above shows that every 2-dimensional non-abelian Lie algebra is isomorphic to the Lie algebra defined by $[x, y] = x$. To calculate the Killing form,

$$\text{ad}_x(x) = 0, \quad \text{ad}_x(y) = x$$

The ad_x is represented by

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

And

$$\text{ad}_y(x) = -x, \quad \text{ad}_y(y) = 0$$

The ad_y is represented by

$$\begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix}$$

It is straightforward to check that

$$\kappa(x, x) = 0, \quad \kappa(y, y) = 1, \quad \kappa(x, y) = 0$$

Thus κ is represented by

$$\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

under the basis $\{x, y\}$.

□

Exercise 4.3

Let L be a Lie algebra over $\mathbb{F}$. Prove that L is solvable if and only if $[LL]$ lies in the radical of the Killing form of L .

Proof. If L is solvable, then it follows from Lie's Theorem that there is a basis

$\{e_1, \dots, e_n\}$ of L in which every element of $\text{ad}(L)$ is represented by an upper triangular matrix.

- Every element of $\text{ad}(L')$ is represented by a strictly upper triangular matrix:

Since e_1 is simultaneous an eigenvector of x and y . Let $L_k = \text{span}\{e_1, \dots, e_k\}$, then

$$z(e_k) = x(y(e_k)) - y(x(e_k))$$

Suppose ,

$$y(e_k) = \sum_{j=1}^k y_j e_j, \quad x(e_k) = \sum_{j=1}^k x_j e_j$$

then

$$\begin{aligned} x(y(e_k)) + L_{k-1} &= x(y_j e_k) + L_{k-1} \\ &= x_j y_j e_k + L_{k-1} \end{aligned}$$

Similarly,

$$y(x(e_k)) + L_{k-1} = x_j y_j e_k + L_{k-1}$$

Thus

$$z(e_k) \in L_{k-1}$$

It follows that z is nilpotent. Thus every element of L' is represented by a strictly upper triangular matrix.

- Take any $x \in L, y \in L'$, The matrix of x and y under the basis $\{e_1, \dots, e_n\}$ are upper triangular matrix and strictly upper triangular matrix, respectively, say A, B . Then

$$\kappa(x, y) = \text{tr}(AB) = \sum_{i=1}^n \sum_{k=1}^n a_{ik} b_{ki} = 0$$

Since for each $i > k$, $a_{ik} = 0$, and for each $i \leq k$, $b_{ki} \leq 0$. That is $[LL]$ lies in the radical of the Killing form of L .

Conversely, if $[LL]$ lies in the radical of the Killing form. We only need to shows that $\text{ad}(x)$ is nilpotent for each $x \in L'$, then it follows from the Engel's Theorem (second version) that L is nilpotent, thus is solvable.

For which, let $\text{ad}(x) = s + n$, where s is semisimple and n is nilpotent, $[s, n] = 0$. We aim to show that $s = 0$. We fix a basis in which s is represented

diagonal and n is represented strictly upper triangular. Note that

$$\operatorname{tr}(\operatorname{ad}(x) \circ S) = \operatorname{tr}(S^2) + \operatorname{tr}(NS) = \operatorname{tr}(S^2) = \sum_i \lambda_i^2$$

where λ_i s are the diagonal entries of S .

There other hand , there are $a_j, b_j \in L$ such that

$$x = \sum_j [a_j, b_j]$$

We have

$$\operatorname{tr}(\operatorname{ad}(x) \circ S) = \sum_j \operatorname{tr}([\operatorname{ad}(a_j), \operatorname{ad}(b_j)] \circ S) = \sum_j \operatorname{tr}(\operatorname{ad}(a_j) \circ [\operatorname{ad}(b_j), S])$$

Since S is a derivation,

$$[\operatorname{ad}(b_j), S] = - \operatorname{ad}(S(b_j))$$

$$\operatorname{tr}(\operatorname{ad}(x) \circ S) = - \sum_j \kappa(a_j, S(b_j))$$

Furthermore, $S(b_j) \in L'$. We have from hypothesis that

$$\kappa(a_j, S(b_j)) = 0, \quad \forall a_j \in L, S(b_j) \in L'$$

Thus

$$\operatorname{tr}(\operatorname{ad}(x) \circ S) = 0$$

There must be $\sum_i \lambda_i^2 = 0$, $S = 0$. Thus $\operatorname{ad}(x)$ is nilpotent, which completes the proof. $\square$

Exercise 4.4

Let V be a finite dimensional vector space over $\mathbb{F}$, equipped with a non-degenerate symmetric bilinear form $\langle \cdot, \cdot \rangle$. Let W be a vector subspace of V , and set

$$W^\perp = \{v \in V \mid \langle v, w \rangle = 0 \text{ for all } w \in W\}.$$

Prove that

$$\dim W + \dim W^\perp = \dim V.$$

Proof. We define a linear map $\Phi : V \rightarrow V^*$, where V^* is the dual space of V , using the non-degenerate bilinear form $\langle \cdot, \cdot \rangle$. For any $v \in V$, $\Phi(v)$ is the linear functional defined by:

$$(\Phi(v))(u) = \langle v, u \rangle \quad \text{for all } u \in V.$$

Since the form $\langle \cdot, \cdot \rangle$ is non-degenerate, the kernel of Φ is $\{0\}$. As V is finite-dimensional, we have $\dim V = \dim V^*$, which implies that the injective map Φ is a linear isomorphism. By the definitions of $W^\perp$ and $W^0 = \{f \in V^* \mid f(w) = 0 \text{ for all } w \in W\}$, we have the following equivalence:

$$v \in W^\perp \iff \langle v, w \rangle = 0, \forall w \in W \iff (\Phi(v))(w) = 0, \forall w \in W \iff \Phi(v) \in W^0.$$

This shows that Φ maps $W^\perp$ isomorphically onto W^0 . Consequently, $\dim W^\perp = \dim W^0$. Using $\dim W + \dim W^0 = \dim V^*$, and the fact that $\dim V = \dim V^*$, we immediately obtain:

$$\dim W + \dim W^\perp = \dim V.$$

This completes the proof. □